

Introduction to R Workshop

Getting Started, Data Visualization, and Regression

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Getting Started

Introduction to R and RStudio

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Getting Familiar with the RStudio Interface

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4. **Files/Plots/Packages/Help/Viewer Panel (Bottom Right):** This panel has multiple tabs for managing files, viewing plots, installing and loading packages, accessing R help documentation, and displaying web content.

Using Markdown

$$E = mc^2$$

Theorem

For all real numbers x and y , we have $(x + y)^2 = x^2 + 2xy + y^2$

Academic Environments

Use `::: {.block}`, `::: {.alertblock}`, `::: {.exampleblock}` for Pandoc-native Beamer blocks. Use raw LaTeX (````\{=latex\}`) for the custom orange/gray environments.

Definition

A **group** $(G, \cdot)$ is a set with a binary operation satisfying closure, associativity, identity, and invertibility.

Proof

By hypothesis every element of $H \leq G$ satisfies $g^n = e$ for some n . Therefore H is generated by g^k for the smallest such k . $\square$

Important

This theorem requires the axiom of choice for infinite collections.

More Environments

Example

The integers $(\mathbb{Z}, +)$ form an abelian group under addition.

Lists and Structure

- First point with important information

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- Second point to demonstrate spacing

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 3. And clean formatting

Two Column Layout

Left Column

- Item one

Right Column

Content on the right side with matching styling

Two Column Layout

Left Column

- Item one
- Item two

Right Column

Content on the right side with matching styling

Two Column Layout

Left Column

- Item one
- Item two
- Item three

Right Column

Content on the right side with matching styling

Code Example

```
# R code with syntax highlighting
library(ggplot2)

ggplot(mtcars, aes(x = wt, y = mpg)) +
  geom_point() +
  theme_minimal()
```

